Chinonso Okorie

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PROFILE

- Operations Research (OR) scientist with 4+ years of experience developing mathematical optimization models and algorithms, along with machine learning (ML) models in transportation industry including transit, ridesourcing, airline, and transportation consultancy.
- Proven track record of implementing production-ready Python, OR, and ML models, with demonstrated computational efficiency and business cost savings.
- Passionate about applying big data analytics, advanced machine learning models, advanced operations research and mathematical modeling to solve complex problems in transportation systems and operations.

EDUCATION

FAMU-FSU College of Engineering, Florida A&M University (FAMU) Tallahassee, FL
PhD candidate in Industrial Engineering – GPA: 4.0 / 4.0 *Expected Graduation – Aug 2027*
Department of Industrial and Manufacturing Engineering

Dissertation: Optimization Models for Charging Infrastructure Planning and Districting in Solid Waste Collection
Advisor: Dr. Yanshuo Sun
Coursework: Integer Programming, Applied Optimization, Intro to Machine Learning, Networked Multi-agent Systems, Big Data Analytics, Deep Learning in Practice, Modeling & Analysis of systems.

FAMU-FSU College of Engineering, Florida A&M University (FAMU) Tallahassee, FL
M.Sc. in Industrial Engineering – GPA: 4.0 / 4.0 *Aug 2022 – Dec 2024*
Department of Industrial and Manufacturing Engineering

Advisor: Dr. Yanshuo Sun

Federal University of Technology (FUTO), Owerri Nigeria
B.Eng. in Mechanical Engineering – Second Class Upper Division, CGPA: 4.1 / 5.0
Department of Mechanical Engineering *Nov 2012 – Feb 2018*
Option of Industrial and Production Engineering

TECHNICAL SKILLS

- Tools & Languages: Python, SQL (Postgres SQL, MySQL, SQLAlchemy), Git & Git version control, Matlab, JAVA, PowerBI.
- Optimization & Mathematical Modeling: GAMS, GAMSPY, IBM ILOG CPLEX, Gurobi, XPRESS, LP, IP, MIP/MINLP, Heuristics, Metaheuristics.
- Machine Learning & Statistical Analysis: Numpy, Pandas, Scikit-Learn, Snowflake, Databricks, Pytorch, Neural Networks, Advanced MS Excel, Minitab for statistical analysis.

CERTIFICATIONS

2024: Web of Science - An Introduction to Peer Review
2023: Udacity - AI Programming with Python
2020: Enipro Data Academy - Proficiency in Data Science & Analytics

WORK EXPERIENCE

Graduate Research Assistant
Industrial and Manufacturing Engineering Department, FAMU-FSU College of Engineering, FL. USA Jan 2023 - Till Date

- Assist in scholarly literature and methodological review of journal papers and technical articles on transportation systems, operations, supply chain and logistics-related research works.
- Develop and implement mathematical optimization models including Integer Linear Programs (ILP), Mixed-integer Programs (MIP), and Stochastic Programs (SP) for transportation and logistics-related research works.
- Develop bi-level MIP model, custom heuristic and exact algorithm for multi-period charging infrastructure planning problem using Tallahassee Community Beautification and Waste Management (CBWM) collection service as a real-world case study.
- Developed MIP models and designed custom solution algorithms for electric-truck route optimization in CBWM collection operations.

- Formulated two network-based optimization models (17+ constraints) for paratransit districting and a shortest-path heuristic for large-scale instances, achieving 15% cost savings and 96% faster computation in live testing.
- Worked with a team of two at the University of Texas at Austin (UT Austin) as part of a multi-university collaboration to investigate the adoption of emerging data-as-a-service (DaaS) for transit demand estimation and service planning across Florida transit agencies.
- Worked with a team of two at the University of North Florida (UNF) as part of a multi-university collaboration to develop an "activation-ratio" metric to quantify mobile ticketing adoption relative to the Jacksonville Transportation Authority (JTA) STAR Card system.
- Built multi-class classification models (Logistic Regression, LightGBM, OVR Logistic Regression) to recommend airline destinations based on real-world bookings data of American Airline, achieving an AUC score of 0.95 and low log loss of 56.7% for dominant classes.
- Designing and conducting experiments, documenting research, reporting findings and manuscript drafting.

Graduate Teaching Assistant
Industrial and Manufacturing Engineering Department, FAMU-FSU College of Engineering, FL. USA Aug 2023 - Till Date

- Assisted in organizing weekly classes for the undergraduate course on OR during the Fall 2023 and Fall 2024 semesters.
- Assisted in organizing weekly classes for an undergraduate course on introduction to data engineering during the spring 2026 semester.
- Graded homeworks and motivation exercises in accordance with the instructor of record's established procedures.
- Organized and maintained a canvas-based gradebook to record student grades and overall performance.
- Monitored class attendance and controlled class activities in accordance with instructor of record's expectations.
- Advised and tutored students during weekly office hours.

PUBLICATIONS & PROJECTS

- Okorie, C. and Sun, Y. (2026). Truck route optimization for electrified municipal solid waste collection: Impacts of network representation and new solution algorithms. Transportation Research Part E: Logistics and Transportation Review, 213:104982.
- Okorie, C., Chen, S., and Sun, Y. (2026). Network-based formulations and solution algorithm for paratransit districting. Transportation Research Part C: Emerging Technologies, 189:105738.
- Okorie, C., Chen, Zhi-Long., and Sun, Y. (2026). Bi-level Optimization for Multi-Period Charging Infrastructure Planning in Waste Collection. IISE Transactions (*Submitted*).
- Enhancing Transit Demand Estimation with Emerging Data-as-a-service (DaaS). Collaborative Project with University of Texas at Austin (UTEXAS), led by FAMU-FSU College of Engineering, Sponsored by FDOT Public Transit Office, Jun 2025 - Till Date.
- JTA Research Lab - Fare Capping Project, public fare affordability, transit accessibility and ridership through fare capping in mobile ticking system. Collaborative Project with University of North Florida (UNF) & FAMU, Sponsored by Jacksonville Transportation Authority (JTA), Dec 2024 - Aug 2025.
- Predictive Modeling of Aircraft Turn Time Pre/Post COVID-19: A Regression Analysis of Delay Impacts using Operational and Flight-Level Data. Finalist Project, FAMU-FSU Big Data Symposium, American Airlines Advanced Analytics Challenge (April 2024).
- Destination Recommender System for Airline Customers Using Multi-Class Classification Models. FAMU-FSU Big Data Symposium, American Airlines Advanced Analytics Challenge (May 2024).

CONFERENCES AND POSTER PRESENTATIONS

- Okorie, C. and Sun, Y. (2026). Network-based Formulations and Solution Algorithm for Paratransit Districting. 2026 Bridging Transportation Researchers (BTR) Conference, Online.
- Okorie, C., Chen, S., and Sun, Y. (2026). Optimization Models for the ADA Paratransit Districting Problem. 2026 Transportation Research Board (TRB) 105th Annual Meeting, Washington D.C.
- Okorie, C. and Sun, Y. (2025). Truck Route Optimization for Electric Trucks in Solid Waste Collection: Impacts of Network Representations and New Solution Algorithms. INFORMS 2025 Annual Meeting, Atlanta, Georgia.

- Okorie, C., Chen, S., and Sun, Y. (2025). Network-based Formulations and Algorithm for the ADA Paratransit Districting Problem. INFORMS 2025 Annual Meeting, Atlanta, Georgia.
- Okorie, C. and Sun, Y. (2025). Route Optimization for Electric Trucks in Solid Waste Collection: Impacts of Network Representations and New Solution Algorithms. 2025 Bridging Transportation Researchers (BTR) Conference, Online.
- Okorie, C. and Sun, Y. (2025). Optimization models for Charging Infrastructure Planning and Districting in Solid Waste Collection. Doctoral Colloquium, 2025 IISE Conference & Expo.

VOLUNTEER & LEADERSHIP EXPERIENCE

Peer Reviewer - Annual Bridging Transportation Researchers (BTR) Conference
Online Transportation Conference Jun 2025 - Till Date

- Reviewed submitted manuscripts for technical soundness and methodological rigor, recommending accept, minor revision, major revision, or reject.
- Provided constructive written feedback to authors on data, methodology, and presentation within assigned deadlines.

Peer Reviewer - Annual Transportation Research Board (TRB) Meeting
National Academies of Science, Engineering & Medicine Aug 2025 - Till Date

- Reviewed submitted manuscripts for technical soundness and methodological rigor, recommending accept, minor revision, major revision, or reject.
- Provided constructive written feedback to authors on data, methodology, and presentation within assigned deadlines.

Chair of Grievance Committee - FAMU Graduate Assistants United (GAU)
Florida A&M University October 2024 - Till Date

- Directing the administration of grievances at FAMU.
- Coordinates the activities of the Grievance Committee as described in the United Faculty of Florida (UFF) constitution and report regularly to chapter.
- Serve as Chapter representative to the State United Faculty of Florida (UFF) Contract Enforcement Committee.

Public Relations Officer - African Graduate Students Association (AGSA FAMU)
Florida A&M University September 2023 - May 2025

- Recruiting AGSA new members.
- Creating and maintaining relations with other RSOs (on and off campus)
- Assisting the communication officer in promoting the organization
- Coordinating the design of promotional materials with the Communication Officer.
- Working with the communication officer to manage the Association's social media account and curate the organization's digital content and engagement.
- Coordinating the posting of promotional materials on campus.

HONORS & AWARDS

- Shirley DeLibero Scholarship for 2026-2027 academic year, American Public Transportation Association (APTA) $7,500.
- Florida Public Transportation Association/Florida Transit Association Finance Corporation (FPTA/FTAFC) Scholarship for 2026-2027 academic year $5,000.
- Corning Endowed Doctoral Fellowship for 2026-2027 academic year, FAMU-FSU College of Engineering $5,000.
- Dr. Yaw Owusu Memorial Scholarship for 2024-2025 academic year, FAMU-FSU College of Engineering $1,250.
- L3Harris Corporate Scholars Fund Scholarship for 2023-2024 academic year, FAMU-FSU College of Engineering $2,000.
- John E. Bates Engineering Excellence Scholarship, FAMU-FSU College of Engineering $500.

PROFESSIONAL MEMBERSHIP

- International Association of Engineers (IAENG) - Member (2026-Present) #582056
- National Society of Black Engineers (NSBE) - APEX member (2022-Present) #467919
- Institute of Industrial and Systems Engineering (IISE) - Graduate Student member #900022787
- Institute for Operations Research and the Management Sciences (INFORMS) - Student Member (2024-Present) #1990903